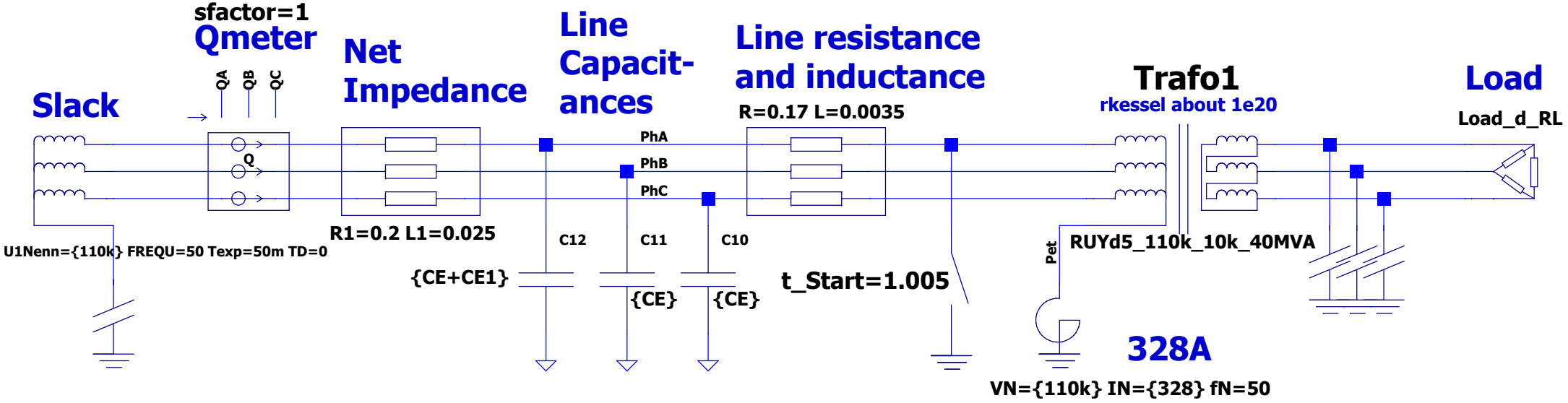
# Middle voltage transformer with petersen IN=328A (TRAN analysis)

```
.param last=20e6
.param last=20e-6
.param tdv=1 tdt=1 ce=5012n ce1=10n fn1=50
.options vntol=1m abstol=1u
.tran 0 5s 0 0.1m
.ac lin 1000 40 52
```




**XC=1/omega/C = 1/314.159/5012e-9=635.1Ohm**
**I0=UN/1.73205/635.1Ohm=100A**
**IE=3*I0=300A (only for cables)**